Effect of Textures and Motion on 3D Shape Perception in Visualization

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INTRODUCTION

To perceive the 3D shape of objects from a two-dimensional image, humans use pictorial cues such as shading and texture. Shading alone, however, does not provide enough detail when viewing a local region of a surface up close. Previous studies have shown that textures aligned with the first principal direction (the direction of greatest curvature at a point on the surface) most improve 3D shape perception (Interrante & Kim 2001).

Motion provides a further strong shape cue. In this project we implemented an algorithm for generating moving principal-direction texture, using two approaches: stroke-based hatching and streamline tracing.

METHODS

Models are stored in a custom .smfd format described as per-vertex positions with precomputed first principal directions, plus a triangle list. The two path-generation methods follow Wang & Kim (2018), whose stroke-based method builds on Girshick et al. (2000). Rendering uses pyglet in Python 3.10.

The two methods differ in where paths are defined. The vertex-based method hops between existing mesh vertices, so paths appear as lines on a mesh. The triangle-based method interpolates directions within triangles using barycentric coordinates and centroids, so paths appear as shorter and curved strokes.

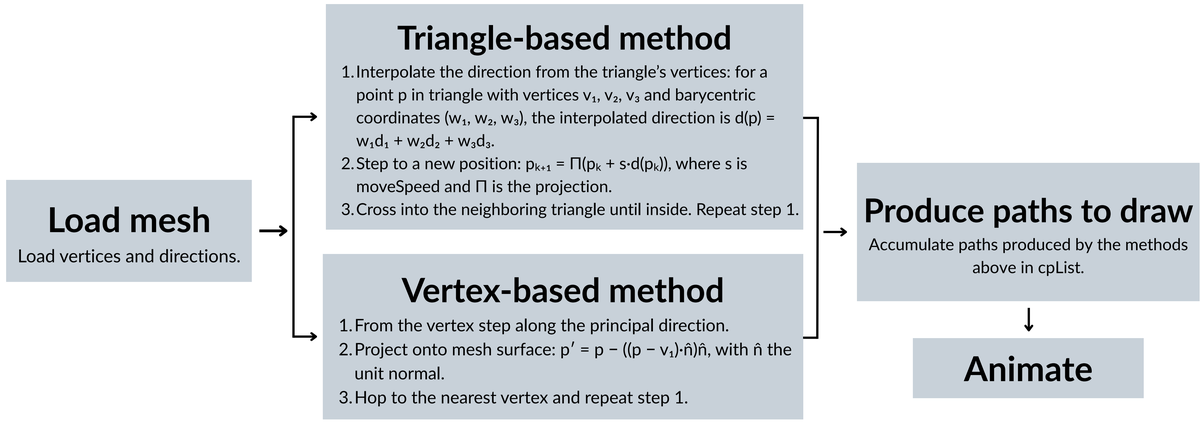


Figure 1. Pipeline for generating principal-direction streamlines.

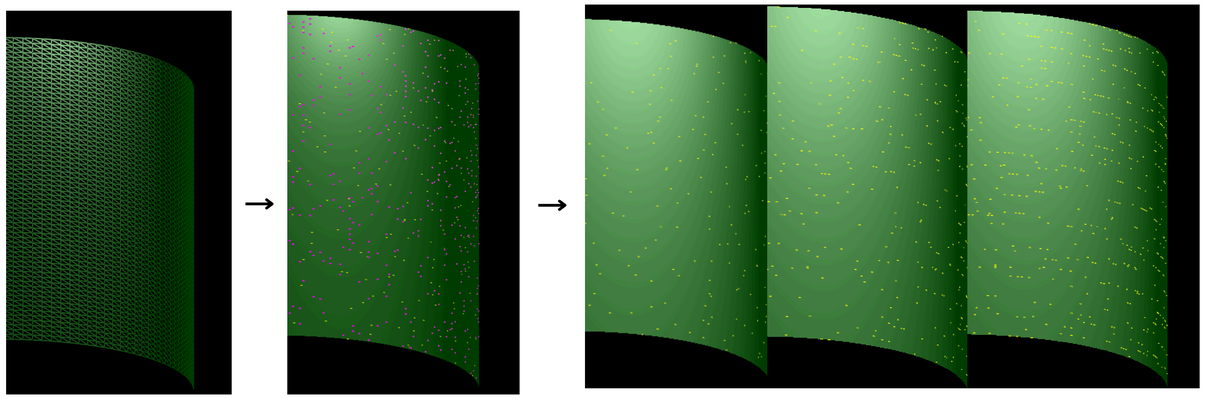


Figure 2. Streamlines advancing over time on a cylinder.

RESULTS & CONCLUSION

The nearest-vertex search in the vertex-based method was a linear scan over all vertices. It was replaced with a KD-tree lookup with O(log n) complexity. Vector operations on 3-element arrays were rewritten as arithmetic to avoid NumPy's per-call overhead and JIT-compiled with Numba.

Mesh	Vertices	Phase	Baseline (s)	Optimized (s)	Speedup
spline_16	110	Vertex paths	0.235	0.068	3.5×
		Triangle paths	1.443	0.08	18×
		Combined	**1.678**	**0.148**	**11×**
spline_25	1,640	Vertex paths	48.61	0.985	49×
		Triangle paths	62.25	2.178	29×
		Combined	**110.86**	**3.163**	**35×**

The improved algorithm produces paths 11× faster on the 110-vertex mesh and 35× faster on the 1,640-vertex mesh. The gain grows with mesh size because the linear search makes path generation quadratic in vertex count, while the KD-tree queries in O(log n). On spline_25 the linear search performed 22 million distance computations, all of which the KD-tree avoids. The vertex-based method improves by 3.5× on the spline_16 mesh and 49× on spline_25, while the triangle-based approach improves by a similar factor on both meshes from JIT compilation and the arithmetic changes.

Both methods project each stepped point onto a triangle plane and get their direction from moving there. If the projection does not fall on the expected triangle, the path direction can be wrong. The original algorithm specified the parameter ptLength = 10. Differences in polygon size can lead to visual artifacts and incorrect direction bending (Figure 3), and whether these become visible depends on the mesh, so the path-length parameter from prior work does not transfer across models and must be set per mesh.

Together these changes move path generation from precomputed to interactive, making it possible to adjust stimuli and view the result directly rather than generating them in advance.

FUTURE WORK

Future work includes replacing the current point sampling and clustering logic with the evenly-spaced placement algorithm of Jacquemet (2024), which guarantees a minimum distance between streamlines. This would replace explicit percentLimit tuning and depth-based clustering with parameters like seed count and radius, where the radius is the minimum distance between streamlines on the surface. The perception study comparing the two motion conditions remains the long-term goal.

REFERENCES

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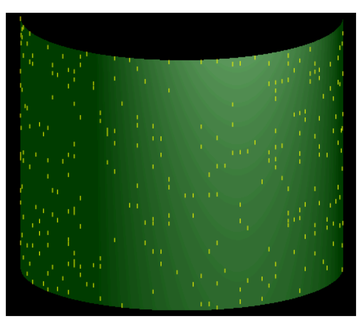


Figure 3. Snapping artifact. Where the projected point lands outside the expected triangle, the stroke takes the wrong direction and bends abruptly.

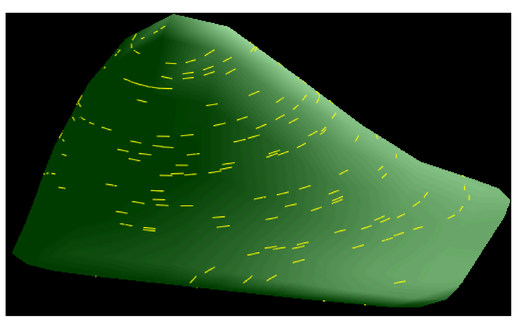


Figure 4. spline_16 (110 vertices) with principal-direction streamlines.

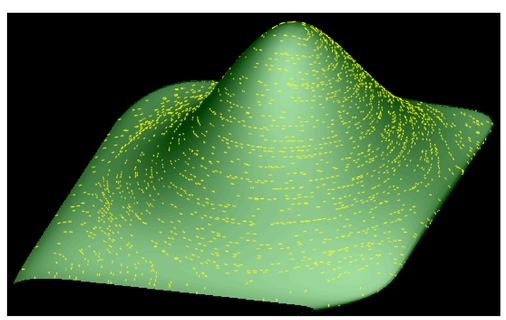


Figure 5. spline_25 (1,640 vertices) with principal-direction streamlines.

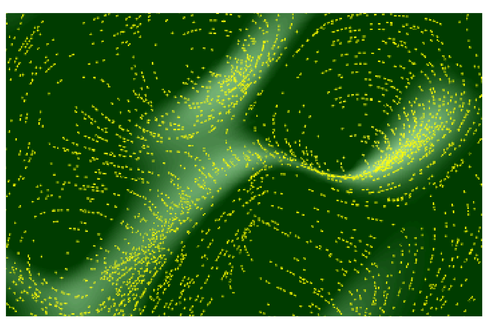


Figure 6. spline_0 (50,562 vertices) with principal-direction streamlines.